

Question 1

What does ADT stand for?

ADT stands for Abstract Data Type.

Question 2

In your own words, what is an Abstract Data Type?

Abstract Data Type defines what something can do without telling how it does it.

Question 3

What is the difference between an ADT and its implementation?

Use the following idea in your explanation:

WHAT versus: HOW

ADT defines what it does and implementation defines how it does it.

Question 4

Can two programmers create different implementations of the same ADT?

Explain your answer.

Yes, because as long as they follow the same ADT rules it does not matter how they are implemented.

Question 5

If one programmer creates a Stack using an array and another creates a Stack using a linked list, are both still Stacks?

Explain why.

Yes, because they both still follow the Stack rules it does not matter which method they use what matter if they follow the same rules last in first out.

Question 6

What does LIFO mean?

LIFO = Last In First Out.

Question 7

Why did 55 get removed before 15?

Stack follow the rule of LIFO because 55 was the last to be put in is the first to get removed.

Question 8

If the Stack contains:

- A
- B
- C
- D

and D was added last, which item should `pop()` remove first?

`pop()` will first remove D.

Question 9

Give one real-world or software example where a Stack could be useful.

Examples discussed in class may include:

- Browser Back history
- Undo operations
- Function calls

Explain your example.

The undo button in a document editor. Everytime you type a word, delete a word, copy and paste a word, bolding a word ... etc gets pushed on to an Undo stack. When you click Undo, the most recent edit is popped off and reversed.

Question 10

What does FIFO mean?

FIFO = First In First Out

Question 11

Why was 15 removed before 55?

Queue follows the rule of FIFO where 15 comes first and it will be the first to come out.

Question 12

If customers enter a line in this order:

Alex

Maria

John

Sarah

Who should leave the Queue first?

Alex

Question 13

Give one real-world or software example where a Queue could be useful.

Possible examples:

- Printer jobs
- Customer-service requests
- Tasks waiting to be processed
- People waiting in line

Explain your answer.

People waiting in line to buy a drink or food in a fast food place.

For each scenario below, choose:

Stack or: Queue

Then explain your answer in one or two sentences.

Scenario 1 — Undo Feature

A text editor remembers your recent actions.

If you type:

A

B

C

the most recent action should be undone first.

Stack or Queue?

Explain.

Stack because undo reverses the most recent action. Typing A, B, C means C gets undone first. Which is LIFO order. A queue would undo A first. Which is the opposite of what you want.

Scenario 2 — Printer

Three students send documents to a printer.

The first document submitted should normally print first.

Stack or Queue?

Explain.

Queue because the first document submitted should print first. This is the FIFO order. Which is exactly how a queue works. A stack would print the last submitted document first. Which would be unfair to whoever sent theirs first.

Scenario 3 — Browser Back Button

You visit:

Google

YouTube

GitHub

Amazon

You click the Back button.

Which page should appear first?

What ADT does this resemble?

If you click the Back button it will show GitHub. Browsing history uses the stack ADT Last In First Out, so the most recently visited page Amazon is popped off first when you hit Back and will show GitHub.

Scenario 4 — Customer Service

Customers are waiting to talk to an employee.

The person who arrived first should normally be helped first.

Stack or Queue?

Scenario 5 — Plates

You place five plates on top of one another.

Which ADT does this represent?

Explain.

Queue because the first customer to arrive should be helped first following the queue ADT of FIFO order, which is exactly how a queue works. A stack would help the most recent arrival first. Which would be unfair to those who have been waiting longest.

Stack

Start with an empty Stack.

push(7)

push(12)

push(18)

pop()
push(22)
peek()

Question 14

What does `pop()` return?

18

Question 15

What does the final `peek()` return?

22

Start with an empty Queue.

enqueue(7)
enqueue(12)
enqueue(18)
dequeue()
enqueue(22)
peek()

Question 16

What does `dequeue()` return?

7

Question 17

What does the final `peek()` return?

12

Feature	Stack	Queue
Rule	LIFO	FIFO
Add operation	push()	enqueue()

Remove operation	pop()	dequeue()
View next item	peek()	peek()
First item removed	The Last item that was placed	The first item that was placed

Question 18

If you implement a Stack using an array, which part is the ADT?

The ADT is just the set of operations and their behaviour not the how. The array is just one way to implement the way it is used to store the data.

Question 19

Which part is the implementation?

Implementation is how it does it. Is the internal mechanics that actually make something work.

Question 20

If you replace the array with a linked list but keep the same Stack operations, did the ADT change?

Explain.

No, it did not because ADT is just the concept of what it needs to do. Changing how it does it does not change the ADT unless you change the behaviour of what it needs to do.